## Quotation No : Q25000315

Date : 12-Aug-2025

Reliance Industries Ltd
Mumbai
NAVI MUMBAI
MAHARASHTRA

## Kind Attention : Mr./Ms. Prashant Shingala

Dear Sir,

We understand that in industry, reliability is not just a requirement,it is the foundation of success.
That is why Absstem Industrial PSA Generator is more than just a product; it is a commitment to excellence, efficiency, and uninterrupted performance.

Every component is designed with precision and innovation, ensuring that each molecule of oxygen and nitrogen is delivered with the utmost purity.

With fully automated operation, our system seamlessly integrates into your processes, reducing manual intervention and maximizing productivity.

Moreover, our industry-leading air ratio ensures that you achieve the highest efficiency, optimizing energy consumption while maintaining peak performance.
And because we believe in standing by our craftsmanship, we offer an unmatched-10 year warranty on sieves, giving you the peace of mind that your investment is protected for the long run.

We have attached the detailed quotation for your review.

Should you have any questions or require further customization to meet your specific needs, please do not hesitate to reach out.

We are here to provide not just a product but a solution that breathes life into your operations.

We look forward to the opportunity to serve you and build a lasting partnership.

Thanks and Regards,

## For Absstem Industrial Solutions Pvt. Ltd.

Pramod Kumar
Director
8447642325
www.absstem.com




nGEN POWERED BY ABSSTEM INDUSTRIAL

# Industrial

# PSA Nitrogen Generator

## Unique Features of Absstem Nitrogen Generator nGEN UH

a. **Lowest Air to Gas ratio** in industry thereby guaranteeing **lowest power consumption**.
b. **10 years*** warranty on Molecular Sieves **(First In Industry)**.
c. Assured Performance for consistent Flow, Pressure and Purity of Nitrogen throughout Product Lifecycle.
d. IIOT ready machine having 100% operator free design for 24X7 operation.
e. Idle mode with auto start/stop features ensuring power savings during low/no demand.
f. Inbuilt dew point transmitter to monitor input compressed air quality as standard scope of supply.
g. Plug n Play machine with an ergonomic design suitable for accessibility.
h. All the Instruments are specially manufactured to make them suitable for Nitrogen gas.
i. All critical performance parameters are being monitored through HMI.
j. First in industry **Comprehensive Maintenance Contract (CMC)** service under Absstem Premium Shield program providing complete peace of mind to customer.

## Ambient Design Parameters

| | | |
|---|---|---|
| a. | Temperature | 0 to 45 Deg. Celsius |
| b. | Relative Humidity | up to 80 % (Max.) |
| c. | Altitude | up to 1000 m above Sea level |

**Absstem Industrial Solutions Pvt. Ltd.**
Plot No. 591, Sector 8, Phase IV
IMT Manesar, Gurugram, Haryana - 122051
Contact - 1800 3010 3394 Email - indsales@absstem.com
CIN: U29256HR2022PTC103387

## Product Description

| | |
|---|---|
| a. Product Name | Ultra Purity PSA Nitrogen Gas Generator |
| b. PSA Module | **nGEN UH** |
| c. Purity Range | 99.99% - 99.999% |

## General PSA Generation System Layout

A typical Industrial Nitrogen Generation system consists of three parts:

- Compressed Air System: Air Compressor, Pre-filter, Air Dryer, Dry Air Vessel
- PSA $N_2$ Generation System
- $N_2$ Storage Tank




| | |
|---|---|
| **Air Compressor** | An Air compressor is used to compress the atmospheric air to a pressure required for the separation of **Nitrogen** from the air. |
| **Refrigeration dryer** | Compressed air then passes through a refrigerated dryer to remove moisture (which is harmful to sieves) from the air. |
| **Air Tank** | Dry air is collected in an air tank to remove the pulsations which serve the dynamic (changing) requirement of air by the PSA system. |
| **Multistage Filtration system** | Before air enters the PSA system, it is passed through multistage filtration removing moisture, particulates and oil content to the specified levels. |
| **PSA system** | Pressurized air is passed through PSA Towers filled with Molecular sieves to separate **Nitrogen** from Air. |
| **Storage Tank** | Industrial Grade **Nitrogen** produced from the PSA System is stored in this for continuous supply. |
| **Oxygen Analyser** | Nitrogen produced is monitored by an Oxygen analyser. In case, $O_2$ impurity goes above the specified level, it gives an alarm and switches the plant to vent mode. |

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## HD Touch Screen Display

(User-friendly touch screen interface for easy operation and maintenance)

Absstem industrial Nitrogen generators complies with all industry standards and regulation

- Alarm management system
- Process data display
- Consumables running hours' display
- Data logging via USB drive**
- Graphical representation**




## Additional upgrades

- Digital flow meter with totalizer
- Nitrogen Pressure monitoring for High pressure booster driven plants
- Cylinder manifold and alarm management system
- Remote monitoring
- Nitrogen gas Dew-point monitoring

## International Quality standards

A strict quality procedure, following the leading international pharmacopoeia and Industrial directives, is followed in the selection of all mechanical and electrical instruments, vessel fabrication and designing of Industrial Gas generators.

TÜV NORD ISO 9001 · TÜV NORD ISO 14001 · TÜV NORD ISO 13485 · RIR CERTIFICATION Partners in Growth



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# Absstem Shield Premium

## (Comprehensive Annual Maintenance Contract)




Any Industry is now heavily focused on increased OEE (Overall Equipment Efficiency) and reduced LCC (Life Cycle Cost) while deciding to invest in any equipment. At Absstem, we strive for an everlasting engagement with our customers by not selling them just a product but also provide a long term after-sales support.

We understand that to deliver a consistent performance, periodic health check and timely maintenance of the Nitrogen Generator is extremely important. With Absstem Shield Premium, you invest in complete peace of mind for all maintenance needs pertaining to N2 generators and associated integrated equipment like air compressor, Air dryer etc.

Few benefits of Absstem Shield Premium are listed as below: -

- Guaranteed assurance of plant performance.
- Genuine spare parts and consumables being used to provide longevity and durability.
- Regular visits by our technicians for health monitoring and carrying out the scheduled maintenance services.
- 24 x 7 On-Call service support to our customers for any basic trouble shooting
- Emergency visit to attend any unplanned breakdown to any location within 48 hours.
- No additional surprise charges as all breakdown spares and consumables covered.
- The contract can be signed at the time of equipment ordering itself and can be renewed on year on year basis.



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# Product Description

| Product | Description | Make | Qty |
|---|---|---|---|
| Compressed Air System | C090F0600<br>Air System | | 8 |

| Caption | Specification |
|---|---|
| Air Compressor KW | 90 KW |
| Air Compressor Make | Atlas Copco/ Ingersoll Rand/ Gardner Denver/ As per Demand |
| Air Dryer | Included |
| Dry Air Storage | 5000 Liters Included |
| Interconnecting Piping | Included |
| Pre filter | Included |
| Dry Air Generated | 602 CFM |
| Air Pressure | 7.5 Barg |
| POWER SUPPLY | 400 V 3 Phase 50 Hz |

| Product | Description | Make | Qty |
|---|---|---|---|
| nGen Nitrogen Generator Ultra Purity | nGEN110UH<br>Nitrogen | | 8 |

| Caption | Specification |
|---|---|
| Gas Generated | Nitrogen |
| Gas Pressure | upto 6.5 Barg |
| FLOW AT 99.99% PURITY | 172 Nm3/Hr DRY AIR REQUIRED 486 CFM |
| FLOW AT 99.995% PURITY | 154 Nm3/Hr DRY AIR REQUIRED 545 CFM |
| FLOW AT 99.999% PURITY | 116 Nm3/Hr DRY AIR REQUIRED 546 CFM |
| COMPRESSED AIR PRESSURE REQUIRED | 7.5 Barg |
| RECOMMENDED NITROGEN STORAGE | 4000 Liters |
| POWER SUPPLY | 230 V/ 110 V 50/60 Hz Single Phase |
| RECOMMENDED AIR RECEIVER | 4000 Litres Part of Air System |
| INPUT AIR DEW POINT MONITOR | INCLUDED |
| CMS Warranty | 10 Years* |



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**CIN**: U29256HR2022PTC103387

| Product | Description | Make | Qty |
|---|---|---|---|
| Gas Storage Vessel | AI-4000L<br>Storage Vessel | | 8 |

| Caption | Specification |
|---|---|
| Water Volume | 4000 Litres |
| Gas Stored | Nitrogen/Oxygen |
| Design Standard | ASME Sec. VIII Div.1 Pressure Vessel Design |
| Working Pressure | 8 Barg |
| Hydrotest Pressure | 13.2 Barg |
| Accessories Included | Pressure Guage Safety Valve Manual Drain Additional Port for Other Instruments |
| Material of Construction | IS2062 Gr.B |
| Gas Stored Volume | 26 M3 |
| Corrosion Allowance | 1.5 MM |

## Commercial Offer

| S No | Item Code | Item Name | Description | HSN | Qty | Unit Price | GST(%) | Amount |
|---|---|---|---|---|---|---|---|---|
| 1 | ACSC090F | C090F0600 | Compressed Air System | 840510 | 8 | 34,15,000.00 | 18.00 | 2,73,20,000.00 |
| 2 | nGEN110UH | nGEN110UH | nGen Nitrogen Generator Ultra Purity | 840510 | 8 | 84,29,000.00 | 18.00 | 6,74,32,000.00 |
| 3 | AIS0408ST | AI-4000L | Gas Storage Vessel | 731100 | 8 | 2,80,000.00 | 18.00 | 22,40,000.00 |
| | | | | | | | Sub Total | 9,69,92,000.00 |
| | | | | | | | GST(%) @18.00 | 1,74,58,560.00 |
| | | | | | | | Grand Total () | 11,44,50,560.00 |



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# Terms And Conditions

| | | |
|---|---|---|
| 1. | EXCLUSIONS & CLIENTS SCOPE | All erection / civil work, All Power Cables & Cabling out of battery limit of PSA Plant, Cable Trenches etc. Unloading / Shifting of Gas Plant at Clients site will be in Client Scope. Additional gas piping, cooling water (if required). |
| 2. | SUPERVISION OF ERECTION & COMMISSIONING | There shall be no commissioning charges, however you shall arrange freeTo & Fro Air Ticket, lodging / boarding, and local conveyance for our Engineers during supervision of Installation of Gas Plant at site on free of cost basis. Our Engineers will be there for Supervision only. Any manpower / fitter required for installation work shall be provided by you. |
| 3. | Freight | Extra at Actuals. |
| 4. | Delivery | 10-12 Weeks from the PO and Advance. Rest can be confirmed for Availability. |

Hope you will find our offer in line with your requirement.

Should you require any further clarification/information, please feel free to contact us.

Thanking you, and assuring you of our best attention at all times.

Very Yours truly,

For Absstem Industrial Solutions Pvt. Ltd.

Pramod Kumar

Director

8447642325

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